1. Hospital Emergency Room Triage

A hospital uses the following triage system for patients: If the patient's temperature is above 102°F or they report severe pain, classify them as "High Priority." If the temperature is between 99°F and 102°F or they report moderate pain, classify them as "Medium Priority." Otherwise, classify them as "Low Priority." Take the patient's temperature and pain level ("none," "moderate," or "severe") as input, and display their priority.

In [41]:
```
temp=input("Please enter the temperature in degree °F: ").replace(" ","").split("°")#if the user puts ° in input by mistake
c=["severe","moderate","none"]
d=temp[0]
if d.replace(".","").isnumeric():
    pain=input("""Please enter the pain level of the patient:
    Severe/Moderate/None: """).lower().strip()
    if pain in c:
        d=float(d)
        if d>102 or pain=="severe":
            print("High Priority")
        elif 99<=d<=102 or pain=="moderate":
            print("Medium Priority")
        else:
            print("Low Priority")
    else:
        print("Please Enter a valid Pain Level")
else:
    print("Please enter a valid Temperature ")
```

```
High Priority
```

2. Shopping Cart Free Delivery

A store provides free delivery under the following conditions: If the total cart value exceeds ₹2000. If the total cart value is between ₹1000 and ₹2000 but the user has a membership. Otherwise, the delivery fee is ₹100. Take the cart value and membership status ("yes" or "no") as input, and determine if the delivery is free or chargeable.

In [43]:
```
cart_value=input("Please enter the cart value(amount only): ₹").lower().replace(" ","")
if cart_value.replace(".","",1).isnumeric():
    cart_value=float(cart_value)
    print(cart_value)
    if cart_value>2000.00:
        print("Eligible for Free delivery")
```

```
    elif 1000.00<=cart_value<=2000.00:
        mem_sta=input("Do you have a membership: Please enter Yes or No: ").lower().strip()
        if mem_sta in ["yes","no"]:
            if mem_sta=="yes":
                print("Eligible for Free delivery")
                # print("Please pay: ₹",cart_value)
            else:

                print("Delivery is chargeable: ₹100")
                # print("Total Cart Value is: ₹",cart_value)
                # print("Total amount is rupees= ₹",cart_value+100.00)
        else:
            print("Please enter a valid member status")
    else:
        print("Delivery is chargeable: ₹100")
        # print("Total Cart Value is: ₹",cart_value)
        # print("Total amount is rupees= ₹",cart_value+100.00)
else:
    print("Please enter valid amount")
```

```
1600.0
Eligible for Free delivery
```

3. Library Fine System

A library charges a fine for overdue books as follows: Up to 5 days overdue: ₹2 per day. 6–10 days overdue: ₹5 per day. More than 10 days overdue: ₹10 per day Take the number of overdue days as input and calculate the fine

In [45]:
```
fine=0
overdue=input("Please enter the number of days overdue(numbers only):").lower().replace(" ","")
overdue=overdue.split("d")# if the user put days in input by mistake
if overdue[0].isnumeric():
    overdue=int(overdue[0])
    if overdue<=5: #up-to 5 days overdue
        fine=overdue*2
    elif 6<=overdue<=10: #6-10 days overdue
        fine=overdue*5
    else: #more than 10 days
        fine=overdue*10
    print(f"You have {overdue} days overdue so total fine is: ₹{fine}")
else:
    print("Please enter a valid number of days Overdue.")
```

```
You have 5 days overdue so total fine is: ₹10
```

4. Airport Luggage Check

Airlines allow the following luggage limits: Economy Class: Up to 20 kg. Excess luggage is charged ₹200 per kg. Business Class: Up to 30 kg. Excess luggage is charged ₹150 per kg. First Class: Up to 40 kg. Excess luggage is charged ₹100 per kg. Take the class type ("economy," "business," or "first") and the luggage weight as input, and calculate the excess fee if applicable.

In [6]:
```
flight_class=input("Please enter the class type: \"Economy\" or \"Business\" or \"First\": ").replace(" ","").lower()
if flight_class in ["economy","business","first"]:
    lug_wei=input("Please enter the luggage weight(in digits please): ").replace(" ","").lower()
    if lug_wei.isalpha():
        print("Please enter valid Luggage weight")
    elif lug_wei.replace(".","").isnumeric():
        # if "k" in lug_wei:
        #     h=lug_wei.split("k")
        #     total_weight=float(h[0])
        # else:
        total_weight=float(lug_wei)

        if flight_class=="economy":
            if total_weight<=20.00:
                print("Flight class: Economy, No extra charges for luggage.")
            else:
                extra= (total_weight-20.00)*200
                print(f"Flight Class: Economy, Extra charges for luggage: ₹{extra}")

        elif flight_class=="business":
            if total_weight<=30.00:
                print("Flight class: Business, No extra charges for luggage.")
            else:
                extra= (total_weight-30.00)*150
                print(f"Flight Class: Business, Extra charges for luggage: ₹{extra}")
        elif flight_class=="first":
            if total_weight<=40.00:
                print("Flight class: First, No extra charges for luggage.")
            else:
                extra= (total_weight-40.00)*100
                print(f"Flight Class: First, Extra charges for luggage: ₹{extra}")
    else:
        print("Please enter the valid luggage weight")
else:
    print("Please enter valid details")
```

```
Please enter valid details
```

5. Hotel Room Pricing

A hotel offers discounts based on the type of customer and room type: Regular customers get 10% off on all rooms. Members get 20% off on deluxe rooms and 15% off on standard rooms. VIP customers get 30% off on all rooms. Take customer type ("regular," "member," or "VIP") and room type ("standard" or "deluxe") as input, and calculate the final price after applying the discount.

In [48]:
```
cust_type=input("Please enter the customer type: \"Regular\", \"Member\",\"VIP\": ").lower().strip()
deluxe_price=5000
standard_price=3500

if cust_type in ["regular","member","vip"]:
    room_type=input("Please enter the room type: \"Standard\", \"Deluxe\":" ).lower().strip()
    if room_type in ["standard","deluxe"]:
        if cust_type=="regular":
            if room_type=="deluxe":
                fin_price=deluxe_price-(deluxe_price*0.1)

            else:
                fin_price=standard_price-(standard_price*0.1)


        elif cust_type=="member":
            if room_type=="deluxe":
                fin_price=deluxe_price-(deluxe_price*0.2)

            else:
                fin_price=standard_price-(standard_price*0.15)

        elif cust_type=="vip":
            if room_type=="deluxe":
                fin_price=deluxe_price-(deluxe_price*0.3)

            else:
                fin_price=standard_price-(standard_price*0.3)

        print(f"Final Price for \"{cust_type.upper()}\" customer in a \"{room_type.title()}\" room type is ₹{fin_price}")

    else:
        print("Please enter a valid Room Type.")

else:
    print("Please enter a valid Customer Type.")
```

```
Final Price for "VIP" customer in a "Standard" room type is ₹2450.0
```

6. Energy Consumption Billing

An electricity company charges based on energy consumption as follows: For consumption up to 100 units, the rate is ₹5 per unit. For consumption between 101 and 300 units, the rate is ₹10 per unit for additional units. For consumption above 300 units, the rate is ₹15 per unit for additional units. Take the total consumption as input and calculate the total bill.

In [50]:
```
consumption=input("Please enter the total units consumed(in digits only): ").lower().replace(" ","")
if consumption.replace(".","",1).isnumeric():
    consumption=float(consumption)
    if consumption<=100:
        tot_price=consumption*5
        print(f"Total amount for {consumption} units is ₹{tot_price}")
    elif 101<=consumption<=300:
        tot_price=500+((consumption-100)*10)
        print(f"Total amount for {consumption} units is ₹{tot_price}")
    else: #applied a programming maths: (upper limit- lower limit)+1
        tot_price=500+2000+((consumption-300)*15)
        print(f"Total amount for {consumption} units is ₹{tot_price}")
else:
    print("Please enter valid Units of consumption")
```

```
Total amount for 25.0 units is ₹125.0
```

7. Ride Fare Calculation

A ride-sharing app charges based on the time of day and distance: Daytime (6 AM to 10 PM): ₹15 per km. Nighttime (10 PM to 6 AM): ₹20 per km. If the distance is more than 20 km, apply a 10% discount on the total fare. Take the time ("day" or "night") and distance as input, and calculate the total fare.

In [63]:
```
time=input("Please enter the time: \"Day\" or \"Night\" ").strip().lower()
if time in ["day","night"]:
    distance=input("Please enter the distance(in digits only): ").lower().replace(" ","")
    if distance.replace(".","").isnumeric():
        distance=distance.split("k")
        distance= float(distance[0])

        if time=="day":
            tot_amount=distance*15
            if distance<=20:
                # tot_amount=distance*15
```

```
                print(f"The final fare for {time} time is ₹{tot_amount}")
            else:

                tot_amount=tot_amount-(tot_amount*0.1)
                print(f"The final fare for {time} time is ₹{tot_amount}")

        else:
            tot_amount=distance*20
            if distance<=20:

                print(f"The final fare for {time} time is ₹{tot_amount}")
            else:
                # tot_amount=distance*20
                tot_amount=tot_amount-(tot_amount*0.1)
                print(f"The final fare for {time} time is ₹{tot_amount}")
    else:
        print("Please enter a valid distance")
else:
    print("Please enter a valid time")
```

```
The final fare for day time is ₹341.955
```

8. Event Entry Rules

For an event, the following entry rules apply: Visitors under 12 or above 60 get free entry. Students (12–25 years) with a valid ID get a 50% discount. All other visitors must pay full price. Take the age and student ID status ("yes" or "no") as input, and determine the entry fee.

In [55]:
```
age=input("Please enter your age: ").replace(" ","").lower().split("y")

# print(age)
if age[0].isnumeric():
    age=int(age[0])
    if age<12 or age>60:
        print("Free entry")
    elif 12<=age<=25:
        status=input("Do you have a student ID? Please enter Yes or No: ").strip().lower()
        if status=="yes":
            print("50% discount applied on valid student ID")
        elif status=="no":
            print("You do not have a valid student ID, please pay the full price")
        else:
            print("Please enter a valid student ID status")
    else:
        print("Please pay the full price")
```

```
else:
    print("Please enter a valid age")
```

```
Please pay the full price
```

9. Parking Lot Charges

A parking lot charges based on the type of vehicle: Cars: ₹50 for the first hour, ₹20 for every additional hour. Bikes: ₹20 for the first hour, ₹10 for every additional hour. Take the type of vehicle ("car" or "bike") and the parking duration as input, and calculate the parking fee.

In [38]:
```
v_type=input("Please enter the Vehicle type: Car or Bike: ").lower().strip()

if v_type in ["car","bike"]:
    time=input("Please enter the duration of parking time").lower().replace(" ","").split("h")
    time=time[0]
    # print(time)
    if time.replace(".","",1).isnumeric():
        time=float(time)
        if v_type=="car":
            if 0<=time<=1:
                tot_bill=50
                print(f"Total amount payable for or within the first hour is ₹{tot_bill}")
            else:
                tot_bill=50+((time-1)*20)
                print(f"Total amount payable for or within {int(time)+1} hour is ₹{tot_bill}")
        else:
            if 0<=time<=1:
                tot_bill=20
                print(f"Total amount payable for or within first hour is ₹{tot_bill}")
            else:
                tot_bill=20+((time-1)*10)
                print(f"Total amount payable for or within {int(time)+1} hour is ₹{tot_bill}")
    else:


        print("Please enter a valid time")
else:
    print("Please enter a valid Vehicle type.")
```

```
Total amount payable for or within 6 hour is ₹60.0
```

10. Real Estate Loan Eligibility

A bank offers real estate loans based on the following criteria: If the applicant earns more than ₹1,00,000 per month and has no existing loans, they are eligible for a 90% loan-to-value (LTV) ratio. If the applicant earns between ₹50,000 and ₹1,00,000, or has existing loans, they are eligible for a 70% LTV ratio. Otherwise, they are eligible for a 50% LTV ratio. Take the monthly income and loan status ("yes" or "no") as input, and display the eligible LTV ratio.

In [56]:
```
income=input("Please enter your Monthly Income(only amount): ₹").replace(" ","").replace(",","")
if income.replace(".","",1).isnumeric():
    income=float(income)
    loan_stat=input("Do you have any existing/ongoing loans?: Yes or No ").strip().lower()
    if loan_stat in ["yes","no"]:
        if income>100000 and loan_stat=="no":
            print("Eligible for a 90% LTV ratio")
        elif 50000<=income<=100000 or loan_stat=="yes":
            print("Eligible for a 70% LTV")
        elif income<50000:
            print("Eligible for 50% LTV ratio")
    else:
        print("Please enter a valid loan status")
else:
    print("Please enter a valid Monthly Income")
```

```
Eligible for 50% LTV ratio
```